▶ PATHOPHYSIOLOGY OF IMPORTANT DISEASES

CONDITION	MECHANISM	PAGE
Lesch-Nyhan syndrome	Defective HGPRT → ↑ de novo purine synthesis → ↑ uric acid production	35
Lynch syndrome (HNPCC)	Mutation of MMR genes (*MLH1*, *MSH2*) → Failure of mismatch repair during the S phase → microsatellite instability → CRC (80%)	37, 395
β-thalassemia	Mutation at splice site or promoter sequences on chromosome 11 → ↓ or absent β-globin synthesis	38, 425
Osteogenesis imperfecta	Mutation in *COL1A1* and *COL1A2* genes → Type 1 collagen defect → inability to form triple helices	49
Ehlers-Danlos syndrome	Faulty collagen synthesis → hyperextensible skin, hypermobile joints, easy bruising	49
Menkes disease	Defective ATP7A protein → impaired copper absorption and transport → ↓ lysyl oxidase activity → ↓ collagen cross-linking	49
Marfan syndrome	*FBN1* mutation on chromosome 15 → defective fibrillin-1 glycoprotein (normally forms sheath around elastin)	50
Prader-Willi syndrome	Imprinting or maternal uniparental disomy (25%) → deletion or mutated expression of paternal allele on chromosome 15	56
Angelman syndrome	Imprinting or paternal uniparental disomy (5%) → deletion or mutated expression of *UBE3A* on maternal chromosome 15	56
Cystic fibrosis	ΔF508 deletion in *CFTR* gene on chromosome 7 → impaired ATP-gated Cl^- channel → impaired secretion of Cl^- in lungs/GI tract and reabsorption of Cl^- in sweat glands	58
Duchenne muscular dystrophy	Dystrophin gene frameshift or nonsense mutations → loss of anchoring protein to ECM (dystrophin) → myonecrosis	59
Becker muscular dystrophy	Non-frameshift deletion of dystrophin gene; less severe than Duchenne	59
Myotonic dystrophy	CTG trinucleotide repeat expansion in *DMPK* gene → abnormal expression of myotonin protein kinase → myotonia	59
Fragile X syndrome	CGG trinucleotide repeat in *FMR1* gene → hypermethylation of cytosine residues → ↓ expression	60
Bitot spots in vitamin A deficiency	↓ differentiation of epithelial cells into specialized tissue → conjunctival squamous metaplasia → keratin buildup	64
Wernicke encephalopathy in patient with alcohol use disorder given glucose	Thiamine deficiency → impaired glucose breakdown → ATP depletion worsened by glucose infusion	64
Pellagra in malignant carcinoid syndrome	Tryptophan is diverted towards serotonin synthesis by tumor → B_3 deficiency (B_3 is derived from tryptophan)	65
Kwashiorkor	Protein malnutrition → ↓ plasma oncotic pressure (→ edema), ↓ apolipoprotein synthesis (→ fatty changes in the liver)	69
Lactic acidosis, fasting hypoglycemia, hepatic steatosis in patient with alcohol use disorder	↑ $NADH/NAD^+$ ratio due to ethanol metabolism	70
Aspirin-induced hyperthermia	↑ permeability of inner mitochondrial membrane → ↓ proton gradient and ↑ O_2 consumption (uncoupling) → heat production	76
Hereditary fructose intolerance	Aldolase B deficiency → Fructose-1-phosphate accumulation → ↓ available phosphate → inhibition of glycogenolysis and gluconeogenesis	78

CONDITION	MECHANISM	PAGE
Essential fructosuria	Fructokinase deficiency → ↑ fructose; hexokinase pathway converts fructose to fructose-6-phosphate	78
Classic galactosemia	Galactose-1-phosphate uridyltransferase deficiency → accumulation of toxic substances (eg, galactitol in eye lens → cataracts)	78
Galactokinase deficiency	Galactokinase deficiency → ↑ galactose; aldose reductase converts galactose to galactitol	78
Cataracts, retinopathy, nephropathy, peripheral neuropathy in DM	Excess glucose → sorbitol (via aldose reductase) and sorbitol → fructose (via sorbitol dehydrogenase); lens, retina, kidney, and Schwann cells lack sorbitol dehydrogenase → intracellular sorbitol accumulation → osmotic damage	79
Recurrent *Neisseria* bacteremia	Terminal complement deficiencies (C5–C9) → failure of MAC formation	105
Hereditary angioedema	C1 inhibitor deficiency → unregulated activation of kallikrein → ↑ bradykinin	105
Paroxysmal nocturnal hemoglobinuria	*PIGA* gene mutation → ↓ GPI anchors for complement inhibitors (DAF/CD55, MIRL/CD59) → complement-mediated intravascular hemolysis → ↓ haptoglobin	105
Type I hypersensitivity	Immediate (minutes): antigen crosslinks IgE on mast cells → degranulation → release of histamine, tryptase, leukotrienes Late (hours): mast cells secrete chemokines (attract inflammatory cells and other mediators) → inflammation, tissue damage	110
Type II hypersensitivity	Antibodies bind to cell-surface antigens or ECM → inflammation, cellular destruction, and dysfunction	110
Type III hypersensitivity	Antigen-antibody complexes activate complement → attract neutrophils → release of lysosomal enzymes	111
Type IV hypersensitivity	T cell–mediated (no antibodies involved); APCs activate CD4+ T cells (→ release of cytokines) or CD8+ T cells (→ direct cell cytotoxicity)	111
Acute hemolytic transfusion reaction	Type II hypersensitivity reaction against donor RBCs (usually ABO antigens)	112
X-linked (Bruton) agammaglobulinemia	Defect in *BTK* gene → no B-cell maturation → absent B cells in peripheral blood, ↓ Ig of all classes	114
DiGeorge syndrome	22q11 microdeletion → failure to develop 3rd and 4th branchial (pharyngeal) pouches → absent thymus and parathyroid	114
Hyper-IgM syndrome	Defective CD40L on Th cells → class switching defect	115
Leukocyte adhesion deficiency (type 1)	LFA-1 integrin (CD18) defect → impaired phagocyte migration and chemotaxis	115
Chédiak-Higashi syndrome	*LYST* mutation → microtubule dysfunction → phagosome-lysosome fusion defect	115
Chronic granulomatous disease	NADPH oxidase defect → ↓ ROS, ↓ respiratory burst in neutrophils → ↑ susceptibility to catalase ⊕ organisms	115
Candida infection in immunodeficiency	↓ granulocytes (systemic), ↓ T cells (local)	114, 116
Graft-versus-host disease	Type IV hypersensitivity reaction; HLA mismatch → donor T cells attack host cells	117
Recurrent *S aureus*, *Serratia*, *B cepacia* infections in CGD	Catalase ⊕ organisms degrade H_2O_2 before it can be converted to microbicidal products by the myeloperoxidase system	126

CONDITION	MECHANISM	PAGE
Hemolytic-uremic syndrome	Shiga/Shiga-like toxins inactivate 60S ribosome → ↑ cytokine release	130, 432
Tetanus	Tetanospasmin cleaves SNARE → inhibition of release of inhibitory neurotransmitters (GABA and glycine) from Renshaw cells	130
Botulism	Toxin (protease) cleaves SNARE → ↓ neurotransmitter (ACh) release at NMJ	130
Gas gangrene	Alpha toxin (a phospholipase/lecithinase) degrades phospholipids → myonecrosis, crepitus	131, 136
Toxic shock syndrome, scarlet fever	TSST-1 and erythrogenic exotoxin A cross-link β region of TCR to MHC class II on APCs outside of antigen binding site → ↑↑ IL-1, IL-2, IFN-γ, TNF-α	131
Shock and DIC caused by gram ⊖ bacteria	Lipid A of LPS macrophage activation (TLR4/CD14), complement activation, tissue factor activation	131
Prosthetic device infection by *S epidermidis*	Biofilm production	126, 133
Endocarditis 2° to *S sanguinis*	Dextrans (biofilm) production → fibrin-platelet aggregates bind to damaged heart valves	126, 134
Pseudomembranous colitis 2° to *C difficile*	Toxins A and B damage enterocytes → watery diarrhea	136
Diphtheria	Exotoxin inhibits protein synthesis via ADP-ribosylation of EF-2 → possible necrosis	137
Virulence of *M tuberculosis*	Cord factor activates macrophages (promoting granuloma formation), induces release of TNF-α; sulfatides (surface glycolipids) inhibit phagolysosomal fusion	138
Tuberculoid leprosy	Th1 immune response, mild symptoms	139
Lepromatous leprosy	Predominantly Th2 response, can be lethal	139
Lack of effective vaccine for *N gonorrhoeae*	Antigenic variation of pilus proteins	140
Cystitis and pyelonephritis 2° to *E coli*	Fimbriae (P pili)	143
Pneumonia, neonatal meningitis 2° to *E coli*	K capsule	143
Chlamydiae resistance to β-lactam antibiotics	Lack of classic peptidoglycan due to reduced muramic acid	146
Influenza pandemics	RNA segment reassortment → antigenic shift	166
Influenza epidemics	Mutations in hemagglutinin, neuraminidase → antigenic drift	166
CNS invasion by rabies	Binds to ACh receptors → retrograde transport (dynein)	169
HIV infection	Binds CD4 along with CCR5 on macrophages (early), or CXCR4 on T cells (late)	172
Granuloma	Macrophages present antigens to $CD4^+$ and secrete IL-12 → $CD4^+$ differentiation into Th1 → IFN-γ secretion → macrophage activation	213
Limitless replicative potential of cancer cells	Reactivation of telomerase → maintains and lengthens telomeres → prevention of chromosome shortening and aging	217
Tissue invasion by cancer	↓ E-cadherin function → ↓ intercellular junctions → basement membrane and ECM degradation by metalloproteinases → cell attachment to ECM proteins (laminin, fibronectin) → locomotion → vascular dissemination	217

CONDITION	MECHANISM	PAGE
Persistent truncus arteriosus	Failure of aorticopulmonary septum formation	285, 302
D-transposition of great arteries	Failure of the aorticopulmonary septum to spiral	285, 302
"Tet spells" in tetralogy of Fallot	Crying, fever, exercise → ↑ RV outflow obstruction → ↑ right-to-left flow across VSD	302
Eisenmenger syndrome	Uncorrected left-to-right shunt → ↑ pulmonary blood flow → remodeling of vasculature → pulmonary hypertension → RVH → right-to-left shunting	303
Atherosclerosis	Endothelial cell dysfunction → macrophage and LDL accumulation → foam cell formation → fatty streaks → smooth muscle cell migration, ECM deposition → fibrous plaque → complex atheromas	305
Thoracic aortic aneurysm	Cystic medial degeneration; associated with 3° syphilis	306
Myocardial infarction	Rupture of coronary artery atherosclerotic plaque → acute thrombosis	308
NSTEMI	Subendocardial infarcts	308
STEMI	Transmural infarcts	308
Death within 0-24 hours post-MI	Ventricular arrhythmia	309, 314
Death or shock within 3–14 days post-MI	Macrophage-mediated ruptures: papillary muscle (2–7 days), interventricular septum (3–5 days), free wall (5–14 days)	309, 314
Wolff-Parkinson-White	Abnormal accessory pathway from atria to ventricle bypasses the AV node → ventricles begin to partially depolarize earlier → delta wave; reentrant circuit → supraventricular tachycardia	311
Hypertrophic obstructive cardiomyopathy (HOCM)	Sarcomere protein gene mutation → concentric hypertrophy (sarcomeres added in parallel); death due to arrhythmia	315
Syncope, dyspnea in HOCM	Asymmetric septal hypertrophy, systolic anterior motion of mitral valve → outflow obstruction	315
Hypovolemic shock	↓ preload → ↓ SV → ↓ CO	317
Cardiogenic shock	↓ CO due to left heart dysfunction	317
Obstructive shock	↓ CO due to blockage of vessels/structures outside the heart	317
Distributive shock	↓ SVR (afterload)	317
Rheumatic fever	Antibodies to M protein cross-react with self-antigens; type II hypersensitivity reaction	319
Deep venous thrombosis	Stasis, hypercoagulability, endothelial damage (Virchow triad) → blood clot within deep vein	321
Most common congenital adrenal hyperplasia	21-hydroxylase deficiency → ↓ mineralocorticoids, ↓ cortisol, ↑ sex hormones, ↑ 17-hydroxyprogesterone	339
Euvolemic hyponatremia in SIADH	↑ ADH → water retention → ↓ aldosterone, ↑ ANP, ↑ BNP → ↑ urinary Na^+ secretion	342
Heat intolerance, weight loss in hyperthyroidism	↑ Na^+/K^+-ATPase → ↑ basal metabolic rate → ↑ calorigenesis	344
Myxedema in hypothyroidism	↑ GAGs in interstitial space → ↑ osmotic pressure → ↑ water retention	344
Graves ophthalmopathy	Lymphocytic infiltration, fibroblast secretion of GAGs → ↑ osmotic muscle swelling, inflammation	346

CONDITION	MECHANISM	PAGE
1° hyperparathyroidism	Parathyroid adenoma or hyperplasia → ↑ PTH	349
2° hyperparathyroidism	↓ Ca^{2+} and/or ↑ PO_4^{3-} → parathyroid hyperplasia → ↑ PTH, ↑ ALP	349
Vascular disease in DM	Nonenzymatic glycation of proteins; small vessel hyaline arteriosclerosis; large vessel atherosclerosis	350
Diabetic ketoacidosis	↓ insulin or ↑ insulin requirement → ↑ lipolysis → ↑ free fatty acid oxidation → ↑ ketogenesis	351
Hyperosmolar hyperglycemic state	Hyperglycemia → ↑ serum osmolality, excessive osmotic diuresis	351
Zollinger-Ellison syndrome	Gastrinoma in pancreas or duodenum → recurrent ulcers in duodenum/jejunum, malabsorption	357
Duodenal atresia	Recanalization failure	366
Jejunal/ileal atresia	Disruption of mesenteric vessels (most commonly SMA) → ischemic necrosis of fetal intestine	366
Superior mesenteric artery syndrome	Diminished mesenteric fat → compression of transverse (3rd) portion of duodenum by SMA and aorta	370
Achalasia	Degeneration of inhibitory neurons in myenteric plexus of esophageal wall → failure of LES relaxation	383
Barrett esophagus	Chronic GERD → metaplasia of nonkeratinized stratified squamous epithelium to intestinal epithelium (nonciliated columnar with goblet cells)	385
Acute gastritis 2° to NSAIDs	↓ PGE_2 → ↓ gastric mucosa protection	386
Celiac disease	Autoimmune-mediated intolerance of gliadin (found in wheat) → malabsorption (distal duodenum, proximal jejunum), steatorrhea	388
Fistula formation in Crohn disease	Transmural inflammation	389
Meckel diverticulum	Persistence of the vitelline (omphalomesenteric) duct	391
Hirschsprung disease	Loss of function mutation in *RET* → failure of neural crest migration → lack of ganglion cells/enteric nervous plexuses in distal colon	391
Adenoma-carcinoma sequence in colorectal cancer	Loss of APC (↓ intercellular adhesion, ↑ proliferation) → *KRAS* mutation (unregulated intracellular signaling) → loss of tumor suppressor genes (*TP53, DCC*)	395
Fibrosis in cirrhosis	Occurs via stellate cell production of ECM	374, 396
Reye syndrome	Aspirin ↓ β-oxidation via reversible inhibition of mitochondrial enzymes	398
Hepatic encephalopathy	Cirrhosis → portosystemic shunts → ↓ NH_3 metabolism	399
α_1-antitrypsin deficiency	Liver: misfolded proteins aggregate in hepatocellular ER → cirrhosis; lungs: ↓ α_1-antitrypsin → uninhibited elastase in alveoli → panacinar emphysema	400
Wilson disease	Mutated hepatocyte copper-transporting ATPase (*ATP7B* on chromosome 13) → copper incorporation into apoceruloplasmin, excretion into bile → ↑ serum ceruloplasmin, copper in tissues and urine	402
Hemochromatosis	*HFE* mutation on chromosome 6 → ↓ hepcidin production, ↑ intestinal absorption → iron overload (↑ ferritin, ↑ iron, ↓ TIBC → ↑ transferrin saturation)	402
Gallstone ileus	Fistula between gallbladder and GI tract → stone enters GI lumen → obstruction of ileocecal valve (narrowest point)	403
Acute cholangitis	Biliary tree obstruction → stasis/bacterial overgrowth	403

CONDITION	MECHANISM	PAGE
Acute pancreatitis	Autodigestion of pancreas by pancreatic enzymes	404
Rh hemolytic disease of the newborn	Rh ⊖ mother forms antibodies (maternal anti-D IgG) against RBCs of Rh ⊕ fetus	411
Anemia in lead poisoning	Lead inhibits ferrochelatase and ALA dehydratase →↓ heme synthesis, ↑ RBC protoporphyrin	425
Anemia of chronic disease	Inflammation → ↑ hepcidin → ↓ release of iron from macrophages, ↓ iron absorption from gut	427
G6PD deficiency	G6PD defect → ↓ NADPH → ↓ reduced glutathione → ↑ RBC susceptibility to oxidant stress	428
Sickle cell anemia	Point mutation → substitution of glutamic acid with valine in β chain → low O_2, high altitude, acidosis precipitate sickling (polymerization of deoxygenated HbS) → anemia, vaso-occlusive disease	428
Bernard-Soulier syndrome	↓ GpIb → ↓ platelet-to-vWF adhesion	432
Glanzmann thrombasthenia	↓ GpIIb/IIIa → ↓ platelet-to-platelet aggregation, defective platelet plug formation	432
Thrombotic thrombocytopenic purpura	↓ *ADAMTS13* (vWF metalloprotease) → ↓ degradation of vWF multimers → ↑ platelet adhesion and aggregation (microthrombi formation)	432
von Willebrand disease	↓ vWF → ↓ platelet-to-vWF adhesion, possibly ↑ PTT (vWF protects factor VIII)	433
Factor V Leiden	Arg506Gln mutation in factor V → resistance to degradation by protein C → hypercoagulable state	433
Heparin-induced thrombocytopenia	Type 1: Heparin administration (within 2 days) → mild, transient drop in platelets; not clinically significant Type 2: Heparin administration (5–10 days) → IgG antibodies against heparin-bound platelet factor 4 complex → complex binds and activates platelets → thrombosis, removal by splenic macrophages → ↓↓ platelet count (significant thrombocytopenia)	440
Warfarin-induced skin/tissue necrosis	Warfarin administration → rapid ↓ in protein C levels due to its shorter half-life → procoagulant state due to delay in depletion of other existing clotting factors with longer half-lives → hypercoagulation and microthrombosis → skin/tissue necrosis	433, 441
Axillary nerve injury	Fractured surgical neck or anterior dislocation of humerus → flattened deltoid, arm at side, ↓ shoulder sensation	450
Radial nerve injury (“Saturday night palsy”)	Axilla compression (use of crutches), midshaft humerus fracture, repetitive pronation/supination of forearm → wrist drop, ↓ grip strength	450
Median nerve injury (hand of benediction)	Proximal lesion: supracondylar fracture → loss of wrist flexion and function of LOAF muscles; loss of sensation over thenar eminence, dorsal and palmar aspect of lateral 3½ fingers Distal lesion: carpal tunnel syndrome	450
Ulnar nerve injury	Proximal lesion: fractured medial epicondyle → radial deviation of wrist on flexion Distal lesion: fractured hook of hamate (fall on outstretched hand) → ulnar claw on digital extension, loss of sensation over ulnar digits	450
Erb palsy (waiter’s tip)	Traction/tear of C5-C6 roots on infant’s neck during delivery, or due to trauma in adults	452

CONDITION	MECHANISM	PAGE
Klumpke palsy (claw hand)	Traction/tear of C8-T1 roots on infant's arm during delivery, or on trying to grab a branch in adults	452
Winged scapula	Injury to long thoracic nerve (C5-C7); seen after mastectomy, stab wounds	452
Common peroneal nerve injury	Trauma to lateral leg, fibular neck fracture → foot drop with "steppage gait"	457
Superior gluteal nerve injury	Iatrogenic injury during IM injection in gluteal region → Trendelenburg sign: lesion contralateral to side of hip that drops due to abductor weakness	457
Pudendal nerve injury	Injury during horseback riding or prolonged cycling; can be blocked during delivery at the ischial spine → ↓ sensation in perineal and genital area +/– fecal/urinary incontinence	457
Radial head subluxation (nursemaid's elbow)	Due to sudden pull on arm (in children; radial head slips out of immature annular ligament)	466
Slipped capital femoral epiphysis	Obese young adolescent with hip/knee pain; ↑ axial force on femoral head → epiphysis displaces relative to femoral neck	466
Achondroplasia	Constitutive activation of *FGFR3* → ↓ chondrocyte proliferation → failure of endochondral ossification → short limbs	467
Osteoporosis	↑ osteoclast activity → ↓ bone mass 2° to ↓ estrogen levels, old age, and long-term use of medications like steroids	467
Osteopetrosis	Carbonic anhydrase II mutations → ↓ ability of osteoclasts to generate acidic environment → ↓ bone resorption → dense bones prone to fracture, pancytopenia (↓ marrow space)	468
Osteitis deformans (Paget disease of bone)	↑ osteoclast activity followed by ↑ osteoblast activity → formation of poor quality, fracture-prone bone	468
Osteoarthritis	Mechanical degeneration of articular cartilage → inflammation with inadequate repair, osteophyte formation	472
Rheumatoid arthritis	Autoimmune inflammation → pannus formation, erosion of articular cartilage and bone	472
Sjögren syndrome	Autoimmune reaction → lymphocyte-mediated damage of exocrine glands	474
Systemic lupus erythematosus	Predominantly a type III hypersensitivity reaction with ↓ clearance of immune complexes; hematologic manifestations are a type II hypersensitivity reaction	476
Blindness in giant cell (temporal) arteritis	Ophthalmic artery occlusion	478
Myasthenia gravis	Autoantibodies to postsynaptic nicotinic (ACh) receptors	480
Lambert-Eaton myasthenic syndrome	Autoantibodies to presynaptic calcium channels → ↓ ACh release	480
Albinism	Normal melanocyte number but ↓ melanin production	484
Vitiligo	Autoimmune destruction of melanocytes	484
Atopic dermatitis	Epidermal barrier dysfunction, genetic factors (ie, loss-of-function mutations in the filaggrin [*FLG*] gene), immune dysregulation, altered skin microbiome, environmental triggers of inflammation	485
Allergic contact dermatitis	Type IV hypersensitivity reaction; during the sensitization phase, allergen activates Th1 cells → memory $CD4^+$ and $CD8^+$ cell formation; upon re-exposure → $CD4^+$ cells release cytokines and $CD8^+$ cells kill targeted cells	485

CONDITION	MECHANISM	PAGE
Pemphigus vulgaris	Type II hypersensitivity reaction; IgG autoantibodies form against desmoglein 1 and 3 in desmosomes → separation of keratinocytes in stratum spinosum from stratum basale; Nikolsky sign ⊕	489
Bullous pemphigoid	Type II hypersensitivity reaction; IgG autoantibodies against hemidesmosomes → separation of epidermis from dermis; Nikolsky sign ⊖	489
Spina bifida occulta, meningocele, myelomeningocele, myeloschisis	Failure of caudal neuropore to fuse by 4th week of development	501
Anencephaly	Failure of rostral neuropore to close → no forebrain, open calvarium	501
Holoprosencephaly	Failure of the forebrain (prosencephalon) to divide into 2 cerebral hemispheres; developmental field defect typically occurring at weeks 3–4 of development; associated with *SHH* mutations	501
Lissencephaly	Failure of neuronal migration → smooth brain surface lacking sulci and gyri	501
Chiari I malformation	Downward displacement of cerebellar tonsils through foramen magnum	502
Chiari II malformation	Herniation of cerebellum (vermis and tonsils) and medulla through foramen magnum	502
Dandy-Walker malformation	Agenesis of cerebellar vermis → cystic enlargement of 4th ventricle, which fills the enlarged posterior fossa	502
Syringomyelia	Fluid-filled, gliosis-lined cavity within spinal cord; damages crossing spinothalamic tract fibers → cape-like loss of pain and temperature	502
Gerstmann syndrome	Lesion in the dominant parietal cortex → agraphia, acalculia, finger agnosia, left-right disorientation	524
Hemispatial neglect syndrome	Lesion in the nondominant parietal cortex → agnosia of contralateral side	524
Klüver-Bucy syndrome	Bilateral lesions in the amygdala; seen in HSV-1 encephalitis → disinhibition, including hyperphagia, hypersexuality, hyperorality	524
Parinaud syndrome	Compression of dorsal midbrain (often due to pineal gland tumors) → upward gaze palsy, convergence-retraction nystagmus, light-near dissociation	524, 542
Cerebral edema	Fluid accumulation in the brain parenchyma → ↑ ICP; may be cytotoxic or vasogenic	525
Aphasia	Stroke in dominant (usually left) hemisphere, in either the superior temporal gyrus of temporal lobe (Wernicke; receptive aphasia) or inferior frontal gyrus of frontal lobe (Broca; expressive aphasia)	526, 529
Locked-in syndrome	Stroke of the basilar artery; loss of horizontal, but not vertical, eye movements	526
Lateral pontine syndrome	Stroke of the anterior inferior cerebellar artery	526
Lateral medullary (Wallenberg) syndrome	Stroke of the posterior inferior cerebellar artery	527
Medial medullary syndrome	Stroke of the anterior spinal artery	527
Neonatal intraventricular hemorrhage	Reduced glial fiber support and impaired autoregulation of BP in premature infants → bleeding into the ventricles, originating in the germinal matrix (a highly vascularized layer within the subventricular zone)	527
Epidural hematoma	Rupture of middle meningeal artery, often 2° to skull fracture involving the pterion	528
Subdural hematoma	Rupture of bridging veins; acute (traumatic, high-energy impact, sudden deceleration injury) or chronic (mild trauma, cerebral atrophy, ↑ age, chronic alcohol overuse, shaken baby syndrome)	528

CONDITION	MECHANISM	PAGE
Subarachnoid hemorrhage	Trauma, rupture of aneurysm (such as a saccular aneurysm), or AVM → bleeding	528
Intraparenchymal hemorrhage	Systemic hypertension (most often occurs in the putamen of basal ganglia, thalamus, pons, and cerebellum), amyloid angiopathy, AVM, vasculitis, neoplasm, or secondary to reperfusion injury in ischemic stroke → bleeding	528
Phantom limb pain	Most commonly following amputation → reorganization of 1° somatosensory cortex → sensation of pain in a limb that is no longer present	529
Diffuse axonal injury	Traumatic shearing of white matter tracts during rapid acceleration and/or deceleration of the brain (eg, motor vehicle accident) → multiple punctate hemorrhages involving white matter tracts → neurologic injury, often causing coma or persistent vegetative state	529
Conduction aphasia	Damage to the arcuate fasciculus	529
Global aphasia	Damage to both Broca and Wernicke areas	529
Heat stroke	Inability of body to dissipate heat (eg, exertion) → CNS dysfunction (eg, confusion), rhabdomyolysis, acute kidney injury, ARDS, DIC	530
Migraine	Irritation of CN V, meninges, or blood vessels (release of vasoactive neuropeptides such as substance P, calcitonin gene-related peptide)	532
Parkinson disease	Loss of dopaminergic neurons of substantia nigra pars compacta	534
Huntington disease	Trinucleotide (CAG) repeat expansion in huntingtin (*HTT*) gene on chromosome 4 → toxic gain of function → atrophy of caudate and putamen with ex vacuo ventriculomegaly; ↑ dopamine, ↓ GABA, ↓ ACh → neuronal death via glutamate excitotoxicity and NMDA receptor binding	534
Alzheimer disease	Widespread cortical atrophy, narrowing of gyri, and widening of sulci; senile plaques in gray matter composed of beta-amyloid core; neurofibrillary tangles composed of intracellular, hyperphosphorylated tau protein; Hirano bodies	534
Frontotemporal dementia (Pick disease)	Frontotemporal lobe degeneration → ↓ executive function and behavioral inhibition	535
Vascular dementia	Multiple arterial infarcts and/or chronic ischemia	535
HIV-associated dementia	Secondary to diffuse gray matter and subcortical atrophy in advanced HIV infection	535
Idiopathic intracranial hypertension	↑ ICP, associated with dural venous sinus stenosis; impaired optic nerve axoplasmic flow → papilledema	536
Communicating hydrocephalus	↓ CSF absorption by arachnoid granulations → ↑ ICP, papilledema, herniation	536
Normal pressure hydrocephalus	Idiopathic, CSF pressure elevated only episodically, no ↑ subarachnoid space volume; expansion of ventricles distorts the fibers of the corona radiata → "wobbly, wacky, wet" triad	536
Noncommunicating hydrocephalus	Structural blockage of CSF circulation within ventricular system (eg, stenosis of aqueduct of Sylvius, colloid cyst blocking foramen of Monro, tumor)	536
Ex vacuo ventriculomegaly	↓ brain tissue and neuronal atrophy → appearance of ↑ CSF on imaging	536
Multiple sclerosis	Autoimmune inflammation and demyelination of CNS (brain and spinal cord) → axonal damage	537

CONDITION	MECHANISM	PAGE
Osmotic demyelination syndrome	Rapid osmotic changes, most commonly iatrogenic correction of hyponatremia but also rapid shifts of other osmolytes (eg, glucose) → massive axonal demyelination in pontine white matter	538
Acute inflammatory demyelinating polyneuropathy (subtype of Guillain-Barré syndrome)	Autoimmune destruction of Schwann cells via inflammation and demyelination of motor and sensory fibers and peripheral nerves; likely facilitated by molecular mimicry and triggered by inoculations or stress	538
Charcot-Marie-Tooth disease	Defective production of proteins involved in the structure and function of peripheral nerves or the myelin sheath	538
Progressive multifocal leukoencephalopathy	Destruction of oligodendrocytes 2° to reactivation of latent JC virus infection → demyelination of CNS	538
Sturge-Weber syndrome	Somatic mosaicism of an activating mutation in one copy of the *GNAQ* gene → congenital anomaly of neural crest derivatives → capillary vascular malformation, ipsilateral leptomeningeal angioma with calcifications, episcleral hemangioma	539
Pituitary adenoma	Hyperplasia of only one type of endocrine cells found in pituitary (most commonly from prolactin-producing lactrotrophs)	540
Spinal muscular atrophy	Congenital degeneration of anterior horns due to *SMN1* mutation → defective snRNP assembly → LMN apoptosis	544
Amyotrophic lateral sclerosis	Combined UMN and LMN degeneration; familial form associated with *SOD1* mutation	544
Tabes dorsalis	Degeneration/demyelination of dorsal columns and roots (in 3° syphilis) → progressive sensory ataxia (impaired proprioception → poor coordination)	544
Poliomyelitis	Poliovirus infection spreads from lymphoid tissue of oropharynx to small intestine then to CNS via bloodstream → cell destruction in anterior horn of spinal cord (LMN death)	544
Friedreich ataxia	Trinucleotide repeat (GAA) on chromosome 9 in frataxin gene (iron-binding protein) → impaired mitochondrial function → degeneration of lateral corticospinal tract, spinocerebellar tract, dorsal columns, and dorsal root ganglia	545
Noise-induced hearing loss	Damage to stereociliated cells in organ of Corti → loss of high-frequency hearing first	548
Presbycusis	Aging-related progressive bilateral/symmetric sensorineural hearing loss (often of higher frequencies) due to destruction of hair cells at the cochlear base	548
Cholesteatoma	Abnormal growth of keratinized squamous epithelium in middle ear; 1° from tympanic membrane retraction pocket; 2° from tympanic membrane perforation	548
Ménière disease	↑ endolymph in inner ear → vertigo, sensorineural hearing loss, tinnitus, ear fullness	548
Hyperopia	Eye too short for refractive power of cornea and lens → light focused behind retina	549
Myopia	Eye too long for refractive power of cornea and lens → light focused in front of retina	549
Astigmatism	Irregular or asymmetric curvature of the cornea or lens → different refractive power at different axes	549

CONDITION	MECHANISM	PAGE
Presbyopia	Age-related impaired accommodation, likely primarily due to ↓ lens elasticity	550
Glaucoma	Optic neuropathy → progressive vision loss (peripheral → central), usually with ↑ intraocular pressure	551
Open-angle glaucoma	Associated with ↑ resistance to aqueous humor drainage through trabecular meshwork	551
Angle-closure glaucoma	Anterior chamber angle narrowed or closed; associated anatomic abnormalities (eg, anteriorly displaced lens resting against central iris) → ↓ aqueous flow through pupil → ↑ pressure in posterior chamber → peripheral iris pushes against cornea → drainage pathways obstructed by iris	551
Diabetic retinopathy	Chronic hyperglycemia → ↑ permeability and occlusion of retinal vessels → microaneurysms, hemorrhages (nonproliferative); retinal neovascularization due to chronic hypoxia (proliferative)	552
Hypertensive retinopathy	Chronic hypertension → spasm, sclerosis, and fibrinoid necrosis of retinal vessels	552
Retinal artery occlusion	Central or branch retinal artery blockage usually due to embolism (carotid artery atherosclerosis > cardiogenic); less commonly due to giant cell arteritis	552
Retinal vein occlusion	Central occlusion due to 1° thrombosis; branch occlusion due to 2° thrombosis at arteriovenous crossings	552
Retinal detachment	Neurosensory retina separates from underlying retinal pigment epithelium → choroidal blood supply loss → hypoxia and degeneration of photoreceptors; due to retinal tears (rhegmatogenous) or tractional or exudative (fluid accumulation) (nonrhegmatogenous)	552
Retinitis pigmentosa	Progressive degeneration of photoreceptors and retinal pigment epithelium	552
Papilledema	↑ ICP (eg, 2° to mass effect) → impaired axoplasmic flow in optic nerve → optic disc swelling (usually bilateral)	553
Relative afferent pupillary defect (Marcus Gunn pupil)	Unilateral or asymmetric lesions of afferent limb of pupillary reflex (eg, retina, optic nerve)	554
Horner syndrome	Lesions along sympathetic chain: 1st neuron (pontine hemorrhage, lateral medullary syndrome, spinal cord lesion above T1 like Brown-Sequard syndrome or late-stage syringomyelia); 2nd neuron (stellate ganglion compression by Pancoast tumor); 3rd neuron (carotid dissection)	555
Cavernous sinus syndrome	2° to pituitary tumor mass effect, carotid-cavernous fistula, or cavernous sinus thrombosis related to infection (spreads due to lack of valves in dural venous sinuses)	557
Delirium	Usually 2° to illness (eg, CNS disease, infection, trauma, substance use, metabolic/electrolyte imbalance, hemorrhage, urinary/fecal retention) or medication (eg, anticholinergics)	575
Schizophrenia	Altered dopaminergic activity, ↑ serotonergic activity, ↓ dendritic branching	577
Distal renal tubular acidosis (RTA type 1)	Inability of α-intercalated cells to secrete H^+ → no new HCO_3^- generated → metabolic acidosis	611
Proximal RTA (type 2)	Defective PCT HCO_3^- reabsorption → ↑ urinary excretion of HCO_3^- → metabolic acidosis	611
Hyperkalemic RTA (type 4)	Hypoaldosteronism/aldosterone resistance → ↑ K^+ → ↓ NH_3 synthesis in PCT → ↓ NH_4^+ excretion → metabolic acidosis	611

CONDITION	MECHANISM	PAGE
Nephritic syndrome	Glomerular inflammation → GBM damage → dysmorphic RBCs in urine, hematuria; ↓ GFR → oliguria, azotemia, ↑ renin release	613
Nephrotic syndrome	Podocyte damage → impaired charge barrier → proteinuria; hypoalbuminemia → ↑ hepatic lipogenesis → hypercholesterolemia; antithrombin loss → hypercoagulability; IgG loss → infections	613
Nephritic-nephrotic syndrome	Severe GBM damage → RBCs lost in urine + impaired charge barrier → hematuria + proteinuria	613
Infection-related glomerulonephritis	Type III hypersensitivity reaction with consumptive hypocomplementemia	614
Alport syndrome	Type IV collagen mutation (X-linked dominant) → irregular thinning, thickening, and splitting of GBM → nephritic syndrome	615
Stress incontinence	Outlet incompetence (urethral hypermobility/intrinsic sphincter deficiency) → leak with ↑ intraabdominal pressure (eg, sneezing, lifting)	618
Urge incontinence	Detrusor overactivity → leak with urge to void immediately	618
Overflow incontinence	Incomplete emptying (detrusor underactivity or outlet obstruction) → leak with overfilling	618
Prerenal azotemia	↓ RBF → ↓ GFR → ↑ reabsorption of Na^+/H_2O and urea	620
Intrinsic renal failure	Patchy necrosis → debris obstructing tubules and fluid backflow → ↓ GFR	620
Postrenal azotemia	Outflow obstruction (bilateral)	620
Adnexal torsion	Twisting of ovary/fallopian tube around infundibulopelvic ligament and ovarian ligament → venous/lymphatic blockage; arterial inflow continues → edema → blockade of arterial inflow → necrosis/hemorrhage	643
Preeclampsia	Abnormal placental spiral arteries → endothelial dysfunction, vasoconstriction, ischemia → new-onset HTN with proteinuria	660
Supine hypotensive syndrome	Supine position → gravid uterus compresses abdominal aorta and IVC → ↓ placental perfusion, ↓ venous return	661
Polycystic ovary syndrome	Hyperinsulinemia and/or insulin resistance → altered hypothalamic feedback response → ↑ LH:FSH, ↑ androgens, ↑ rate of follicular maturation → unruptured follicles (cysts), anovulation	662
Functional hypothalamic amenorrhea	Severe caloric restriction, ↑ energy expenditure, and/or stress → altered pulsatile GnRH secretion → ↓ LH, FSH, estrogen	663
Varicocele	↑ venous pressure → dilated veins in pampiniform plexus → enlarged scrotum (“bag of worms”)	669
Methemoglobin	↑ oxidized Hb (Fe^{3+}) 2° to dapsone, local anesthetics, nitrites → ↓ O_2 binding but ↑ cyanide affinity → tissue hypoxia	688
Sarcoidosis-associated hypercalcemia	Noncaseating granulomas → ↑ macrophage activity → ↑ 1α-hydroxylase activity in macrophages → vitamin D activation → ↑ Ca^{2+}	695
ARDS	Alveolar injury → inflammation → capillary endothelial damage and ↑ vessel permeability → leakage of protein-rich fluid into alveoli → intra-alveolar hyaline membranes and noncardiogenic pulmonary edema → ↓ compliance and $\dot{V}/\dot{Q}$ mismatch → hypoxic vasoconstriction → ↑ pulmonary vascular resistance	697
Sleep apnea	Respiratory effort against airway obstruction (obstructive); impaired respiratory effort due to CNS injury/toxicity, CHF, opioids (central); obesity → hypoventilation → ↑ $PaCO_2$ during waking hours	697